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Electron Mobility Calculations and Scattering Effects in GaN Compound Semiconductor

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Abstract*—Compound semiconductors allow processing in electronic devices much faster than silicon-based devices since electrons can travel much faster than their counterparts in silicon-only devices. Electron mobility for n-GaN compound semiconductor is calculated using simulation via relaxation time at different temperatures (up to 400 K) and different doping concentrations considering four elastic scattering mechanisms; namely: neutral impurity-, ionized impurity-, piezoelectric- and acoustic phonon-scattering. It is found that electron mobility of GaN increases as temperature increases for ionized impurity scattering, but it decreases for acoustic deformation and piezoelectric scatterings; while neutral impurity scattering was not affected by temperature changes. At higher temperatures, mobility due to piezoelectric scattering approaches that of deformation scattering and can be neglected. Mobility, at low temperatures, decreases with increasing doping concentrations. At room temperature, mobility also decreases as the n-GaN becomes increasingly doped most likely explained by massive ionized impurity scattering which restricts electron mobility. The low-field mobility of electrons generally decreases when temperature increases. The obtained results are compared with that from the literature exhibiting very good agreements.

***Keywords*— Compound semiconductors, GaN, Electron mobility, scattering mechanisms.**

I. INTRODUCTION

Electronic devices of different types are fabricated from semiconductor elements, especially silicon (Si) and germanium (Ge). Si, however, is widely used in ICs more than Ge due to its distinctive characteristics. The integration rate of LSIs would double every 18 months, meaning that it would double in three years and become 1,000 times denser in 15 years, according to Gordon Moore, who eventually co-founded Intel. This prediction was made in 1965. In the semiconductor and computer industries, the idea was largely embraced, and reality nearly matched Moore's predictions. Semiconductor fabrication has adhered to Moore's law by reducing device dimensions constantly for the last three decades [1-2]. This was largely due to the remarkable developments in lithography. However, this scaling in CMOS devices is facing considerable challenges at different levels, practical and theoretical, as device dimensions go into few nanometers range. To avoid these challenges and find alternative materials other than silicon and germanium, compound semiconductors, either from III-V elements or II-VI elements, are being investigated and commercially produced due to their wide band gaps, especially III-nitrides like AlN, InN and GaN, are being used for blue and UV LEDs, high-temperature, high-frequency applications, etc. As well, III–nitrides semiconductors are presently employed in new cutting-edge classes of electrical and optoelectronic devices such as automatic control applications, computer technologies, computer engineering, telecommunications, renewable energies, and so on. For example, a III-nitride quantum-well transistor is used as a logic device at a 0.5 V low supply voltage with low power dissipation and high-speed operation [3]. In this paper, one of these III–nitrides compound semiconductors is considered, namely GaN. GaN possesses exceptional characteristics that are in demand for applications that require higher speed and power features. Some of these characteristics are high electron mobility and high breakdown electric fields. Because of their light effective masses, electrons in GaN travel faster than those in Si counterparts allowing fast processing applications, up to 100 times faster. Because of these distinctive assets, GaN technology now is being employed in a range of power electronic devices because of these distinguished properties even at fast switching speeds [4-5].

Due to the importance of GaN-based devices in many applications, electron mobility, temperature dependency and scattering effects in n-GaN are investigated in this paper to fully understand and model electrical characteristics of these devices [6-8].

II. MATERIAL AND METHODOLOGY

Charge carriers' mobility, either electrons or holes, and scattering rates in semiconductors are quantities that significantly influences the operation of electronic devices [9-11]. Although measuring mobility values is somewhat straightforward, theoretically predicting it is a bit challenging. Applying an electric field to a semiconductor increases mobility which is specified by the speed of which an electron and/or hole travels through this semiconductor as a result of the field. As the electric field becomes substantial enough, carriers' velocity saturates, especially in low- and high-field mobility cases. Charge carriers in compound semiconductors follow more quickly this increase in the field acquiring higher mobilities which in turn producing faster

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device response [12]. Usually, simple mobility models only consider temperature and doping concentrations effects. This procedure may provide fast simulation and permits initial account of the device modifications and analysis but may be less precise. However, more comprehensive models are required in order to fit experimental results accurately. One of the parameters that must be included in models is the effect of carriers' mobility. Table I summarizes the relevant GaN characteristics and its most popular applications.

TABLE I. Main characteristics of the GaN compound semiconductor.

Semiconductor Compound	Gallium Nitride (GaN)
Energy gap (eV) at 0 K	3.47 (direct)
Effective mass (kg)	0.22 m_0
Mass density (ρ) (Kg/m^3)	6590
Density [g/cm^3]	6.1
Popular Use	Blue LED, Laser diodes, power electronic converters, high temperature devices and radio-frequency power amplifiers.

The methodology used to determine the electron mobility requires the calculation of ionized impurity scattering as a result of electrical forces between the ionized impurity centers and the charge carriers. This scattering is proportional to the number of impurity dopants and the time of interaction where a lower mobility is caused by higher impurity concentrations. Scattering (or collisions) with impurities, however, cause them to lose momentum. The net effect of these microscopic processes leads to total macroscopic mobilities, that are then employed in transport equations, which are proportional to lattice temperature, magnitude of electric field, impurity concentrations and other factors. Usually modeling mobility is divided into main parts, low-field and high-field mobilities. Charge carriers' interaction with lattice phonons may limit carriers' mobility which may dominate at room temperatures. However, for high dopant concentrations and at low temperatures, carriers scattering due to impurities might be the main factor. For these reasons and others, it is essential to model and study electron mobility [13-14].

III. MODEL CONSTRUCTION

Ionized impurity scattering (Brooks–Herring Formula) is proportional to dopant impurity number and interaction time where a lower mobility is caused by higher impurity concentrations. The relaxation time approximation method is used to determine the n-GaN carrier mobility for different temperature ranges. The relaxation time, τ_{ii}, for ionized impurity scattering is calculated as [15]:

$$\frac{1}{\langle\tau_{ii}(\mathcal{E})\rangle} = \frac{z^2e^4N_i}{16\pi\epsilon^2\sqrt{2m^*}}\mathcal{E}^{-\frac{3}{2}}\left[\log\left(1+\frac{8m^*{\lambda_s}^2\mathcal{E}}{\hbar^2}\right) - \frac{\frac{8m^*{\lambda_s}^2\mathcal{E}}{\hbar^2}}{\left(1+\frac{8m^*{\lambda_s}^2\mathcal{E}}{\hbar^2}\right)}\right] \quad (1)$$

Where m^* is electron effective mass, ϵ is the dielectric constant, N_i is the ionized impurities concentration, $\hbar = h/2\pi$, h is Planck's constant, λ_s is the Debye screening length and ε is electron energy.

When replacing electron energy ε by $3k_BT$ and $1/\lambda_s = (nze^2/\epsilon k_BT)^{1/2}$, where n is the carrier concentration, T is the temperature and k_B is Boltzmann constant, the Brooks–Herring formula becomes:

$$\frac{1}{\langle\tau_{ii}(\mathcal{E})\rangle} = \frac{z^2e^4N_i}{16\pi\epsilon^2\sqrt{2m^*}}\mathcal{E}^{-\frac{3}{2}}\left[\log(1+y) - \left(\frac{y}{1+y}\right)\right] \quad (2)$$

$$\text{where: } y = \frac{24\, m^*\epsilon {k_B}^2T^2}{\hbar^2e^2n} \quad (3)$$

Then, the electron mobility caused by ionized impurity scattering as given by Brooks–Herring formula is

$$\mu_{ii} = \frac{e\langle\tau_{ii}\rangle}{m^*} \quad (4)$$

$$\mu_{ii} = \frac{128\sqrt{2}\,\pi^{\frac{1}{2}}\epsilon^2k^{\frac{3}{2}}T^{\frac{3}{2}}}{N_i\, e^3m^{*\frac{1}{2}}\left[\log(1+y) - \frac{y}{1+y}\right]} \quad (5)$$

The acoustic phonon scattering, or the deformation potential scattering, is a result of electron energy change because of lattice deformation. This change of crystal energy is related to volume (V) change, $\Delta(\boldsymbol{r}) = \delta V/V$, and the interaction between the acoustic phonon and the electron. The scattering rate due to this electron-phonon interaction is given by [16]:

$$\frac{1}{\langle\tau_{dp}\rangle} = \frac{D_{ac}^2k_BT}{2\pi\hbar\rho s^2}\int_0^{2k} q^2\frac{m^*}{\hbar^2k\,q}\,dq = \frac{D_{ac}^2k_BT}{2\pi\hbar\rho s^2}\frac{m^*}{\hbar^2k}\int_0^{2k} q\,dq$$

$$= \frac{D_{ac}^2k_BT\,m^*}{\pi\,\hbar^3\rho s^2}k = \frac{(2\,m^*)^{3/2}D_{ac}^2k_BT}{2\,\pi\,\hbar^4\rho s^2}\mathcal{E}^{\frac{1}{2}} \quad (6)$$

Where ρ is the crystal density, s is the average sound velocity and D_{ac} is the deformation potential for electron scattering due to acoustic phonons. The relaxation time, as a result of this acoustic phonon scattering, yields the mobility as:

$$\mu_{dp} = \frac{e}{m^*}\langle\tau_{dp}\rangle \quad (7)$$

$$\mu_{dp} = \frac{2\sqrt{2}\pi^{\frac{1}{2}}\hbar^4\,\rho\, s^2e}{3D_{ac}^2\, m^{*\frac{5}{2}}\,(k_B\,T)^{\frac{3}{2}}} \quad (8)$$

A polarization P is induced when applying a strain S_{kl} to the crystal with no inversion symmetry. This phenomenon is called piezoelectricity causing lattice vibration. This lattice vibrations induces a potential in the crystal and the electrons are scattered as a result of this piezoelectric potential scattering. Then, the scattering rate ω_{pz} (piezoelectric angular frequency) as a result of this piezoelectric potential is obtained by the following relation [17]:

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$$\omega_{pz} = \frac{2\, e^2\sqrt{m^*}k_B T}{4\sqrt{2}\hbar^2}\langle\frac{h_{pz}^{*2}}{\epsilon^{*2}c^*}\rangle\frac{1}{\sqrt{\mathcal{E}}} \times\left[\log\left(1+\frac{8\, m^*\,\mathcal{E}}{\hbar^2 q_s^2}\right) - \frac{1}{1+\frac{\hbar^2 q_s^2}{8\, m^*\,\mathcal{E}}}\right] \quad (9)$$

A factor two appears to take into account absorption and emission, h_{pz}^* is the effective piezoelectric constant and $\epsilon^*$is the effective dielectric constant and q_s is the screening wavenumber, where q_s is obtained by:

$$q_s = \frac{1}{\lambda_s} = \left(\frac{nze^2}{\epsilon k_B T}\right)^{\frac{1}{2}} \quad (10)$$

Taking the simple case of $q_s = 0$, the relaxation time for piezoelectric potential scattering is found as:

$$\frac{1}{\langle\tau_{pz}\rangle} = \omega_{pz} = \frac{e^2\sqrt{m^*}k_B T}{2\sqrt{2}\hbar^2}\langle\frac{h_{pz}^{*2}}{\epsilon^{*2}c^*}\rangle\frac{1}{\sqrt{\mathcal{E}}} \quad (11)$$

Where c^* is elastic constant ($c^* = \rho v^2$), $h_{pz}^* = h_{pz}$ and $\epsilon^* = \epsilon$.

Then, the mobility is given by:

$$\mu_{pz} = \frac{e\langle\tau_{pz}\rangle}{m^*} \quad (12)$$

$$\mu_{pz} = \frac{16\sqrt{2}\,\pi^{\frac{1}{2}}\hbar^2}{3e\ m^{*\frac{3}{2}}\langle\frac{h_{pz}^2}{\epsilon^{*2}c^*}\rangle k_B^{\frac{1}{2}}\,T^{\frac{1}{2}}} = \frac{16\sqrt{2}\,\pi^{\frac{1}{2}}\epsilon^2\hbar^2\,\rho\, s^2}{3e\, {h_{pz}}^2 m^{*\frac{3}{2}}k_B^{\frac{1}{2}}\,T^{\frac{1}{2}}} \quad (13)$$

Where h_{pz} is piezoelectric constant.

At low temperatures, for the case of neutral impurity scattering, electrons and holes occupy the donor and acceptor available states. These states act as neutral impurities and do not add to Coulomb potential scattering. When an electron is captured by a shallow donor state, its wave function spans a wide range and the ground state's effective Bohr radius (a) is substantially bigger than the constant of the lattice. The scattering cross-section is approximated as follows [18-19]:

$$\sigma_n = \frac{20\, a}{k} \quad (14)$$

where k is the electron wave vector.

The mean free path for the electron is also obtained by:

$$l_n = \tau_n v = \frac{\tau_n \hbar k}{m^*} \quad (15)$$

This electron has a scattering probability given by $N_n\sigma_n$, where N_n is the neutral impurity density. These relations result in:

$$\frac{1}{l_n} = N_n\,\sigma_n \quad (16)$$

Then, knowing the cross-section of the scattering for neutral impurity, the average relaxation time is:

$$\frac{1}{\langle\tau_{ni}\rangle} = \frac{\hbar\, k}{m^*}.N_n\,\sigma_n = \frac{20N_n\, a\,\hbar}{m^*} \quad (17)$$

The assumption behind this approximation is that this cross-section is a result of an s-orbit electron. Hence, the mobility caused by the neutral impurity scattering is obtained from:

$$\mu_{ni} = \frac{e\langle\tau_{ni}\rangle}{m^*} \quad (18)$$

$$\mu_{ni} = \frac{e}{20N_n\, a\,\hbar} \quad (19)$$

$$\mu_{ni} = \frac{e^3 m^*}{80\pi N_n \epsilon\,\hbar^3} \quad (20)$$

Finally, the total mobility μ_{total} is calculated numerically using τ_{total}, which is described by the following equation:

$$\frac{1}{\tau_{total}} = \frac{1}{\tau_{ii}} + \frac{1}{\tau_{dp}} + \frac{1}{\tau_{pz}} + \frac{1}{\tau_{ni}} \quad (21)$$

$$\mu_{total} = \frac{e}{m^*}\langle\tau_{total}\rangle$$

$$\mu_{total} = {\mu_{ii}}^{-1} + {\mu_{de}}^{-1} + {\mu_{pz}}^{-1} + {\mu_{ni}}^{-1} \quad (22)$$

Mobility Modeling

Once an electric field is applied to the crystal, electrons and holes accelerate. Collisions or scattering with impurities, however, cause them to lose momentum. This scattering is caused by phonons (lattice vibrations), impurity dopants, material defects and other carriers. These microscopic processes manifest themselves as macroscopic mobilities in the transport equations, where these mobilities are proportional to a number of factors such as: impurity concentration, thermal vibrations of the lattice, electric-field strength and other factors. These considerations are essential to find mobility values which are in accordance with experimental results. Mobility modeling is generally considered as two parts; low-field mobility and high-field mobility [20].

Low-Field Mobility

The low-field electron mobility (μ_{LF}) depends on lattice thermal vibrations and concentrations of impurities. Because scattering at donors, acceptors and other defects affect mobility, it is conventional to describe this low-field mobility in terms of doping density and temperature as follows [21]:

$$\mu_{LF}(N,T) = \mu_{min} + \frac{\mu_{max} - \mu_{min}}{1+\left(\frac{N_i}{N_g}\right)^{\lambda}} \quad (23)$$

$$\mu_{max} = \mu_{max}(300k)(T/300k)^{\theta_1} \quad (24)$$

$$\mu_{min} = \mu_{min}(300k)(T/300k)^{\theta_2} \quad (25)$$

$$N_g = N_{ref}(300k)(T/300k)^{\theta_3} \quad (26)$$

Where μ_{max} is the mobility in intrinsic (undoped) semiconductor (maximum mobility), μ_{min} is the mobility in extrinsic highly doped semiconductor (minimum mobility), N_i is the ionized impurities concentration, N_{ref} is the reference doping concentration, N_g is the impurity concentration at which mobility value decreases to nearly one-half of its maximum value and λ is a calculated value showing how rapidly the mobility improved from μ_{max} to μ_{min}. θ_1, θ_2 and

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θ_3 are predetermined parameters. Table II presents the parameters of the n-GaN.

TABLE II. Material parameters for the n-GaN semiconductor compound.

Material	**n-GaN**
μ_{max} (300 K) ($cm^2/V.s$)	1600
μ_{min} (300 K) ($cm^2/V.s$)	100
N_{ref} (300 K) (cm^{-3})	3 x 10^{17}
Λ	0.7
θ_1	-1.5
θ_2	-0.2
θ_3	1.3
Reference	[20]

High-Field Mobility

Increasing the electric field results in charge carriers gaining energies greater than the ambient thermal energy. This gained energy is transferred to the crystal lattice through emitting optical phonons. The carrier velocity is saturated as a result of this. To take this velocity saturation into mobility modeling, mobility must be lowered. The Caughey and Thomas expression takes this essential aspect into account for the high-field mobility (μ_{HF}) as follows [12].

$$\mu_{HF} = \frac{\mu_{LF}}{\left[1 + \left(\frac{\mu_{LF} E}{v_s}\right)^{\beta}\right]^{\frac{1}{\beta}}} \qquad (27)$$

Where v_s is the saturation velocity and β is a parameter affecting the transition from low to high field cases. Equation (27) is not just restricted to uniform fields; but it can also be employed in non-uniform field device modeling.

IV. RESULTS AND DISCUSSION

The results of calculating electron mobility for the GaN material are presented. The electron mobility was calculated using the semiconductor characteristics at various temperatures as well as for the various scattering processes. Again, these processes are neutral impurity, ionized impurity, acoustic phonon and piezoelectric scatterings. The ionized impurity concentration, carrier density and neutral impurity concentrations are set to 6 x 10^{20} m^{-3}, 6 x 10^{10} $m^{-3}$and 6 x 10^{21} m^{-3}, respectively. Fig. 1 presents the dependency of mobility change on temperature in the range from 10 K to 400 K for the n-GaN.

From this figure, it is noticed that the ionized impurity mobility increases as the temperature increases, whereas the mobility of acoustic lattice vibrations decreases when the temperature increases. The mobility resulting from the piezoelectric potential scattering is seen to be smaller than that for the deformation potential mobility. At higher temperatures, the curve of mobility due to piezoelectric potential approaches the curve of deformation potential scattering. This suggests that the influence of the piezoelectric acoustic phonons may be ignored for higher temperatures. The mobility related to the neutral impurity scattering process, as expected, is constant (μ_{ni}=5.181$m^2/V.s$) since it is temperature independent. The obtained results are found to be in good agreement with that found in the literatures, e.g. [18].

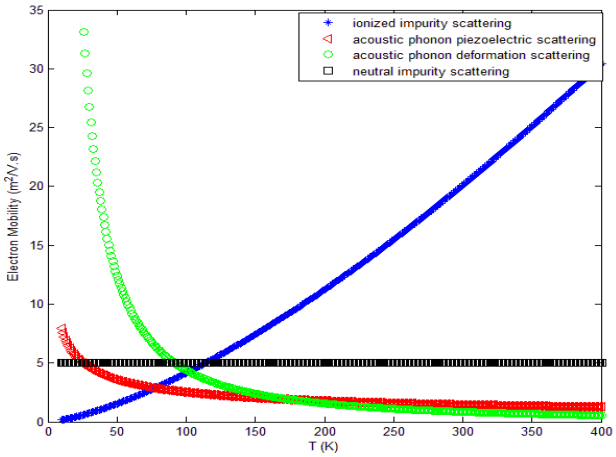

Fig. 1. Dependecy of electron mobility on the temperature for the n-GaN for the discussed scattering processes.

Fig. 2 presents the relation of the electron drift mobility with temperature for the range of 10 $K$to 400 Kat different doping concentrations (6x10^{20} cm^{-3}, 10x10^{20} cm^{-3} and 15x10^{20} $cm^{-3}$and 20x10^{20} cm^{-3}) for n-GaN. The electron concentration is considered to be equal to concentration of impurities and the screening by free carriers is taken into account.

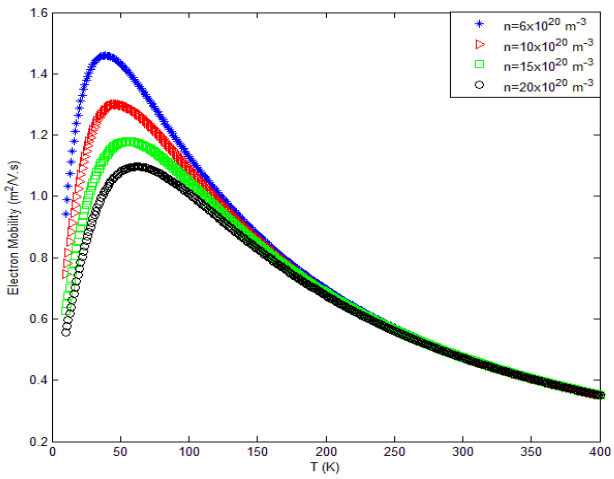

Fig. 2. Dependecy of electron mobility on temperature range up to 400 K for different doping concentrations for n-GaN.

It can be noticed from this figure that the mobility increases as temperature rises until it reaches a maximum value (at low temperatures), after which it decreases (at high temperatures). When the concentration of impurities increases, the peak value of mobility falls. The obtained results are found to be in good agreement with that obtained by other researchers e.g. [14, 18].

Number of charge carriers, either electrons or holes, is controlled by the impurity concentrations and, thus, doping concentrations have a noticable influence on carrier mobility. Hence, Fig. 3 presents the drift mobility dependency on the

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doping concentrations at 100 K, 200 K, 300 K and 400 K. Fig. 3 illustrates that the electron mobility diminishes as the doping in n-GaN increases. This drop in the mobility is most likely explained by massive ionized impurity scattering. This scattering restricts the mobility at low temperatures. The obtained results are found to be in good agreement with those achieved by other researchers e.g. [15, 18].

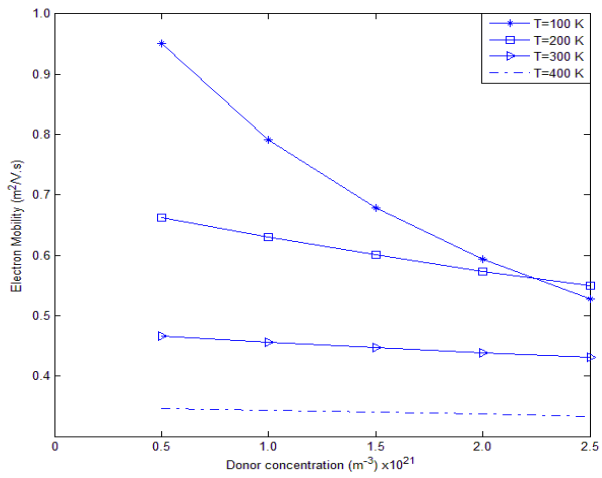

Fig. 3. Dependecy of electron drift mobility on the doping concentrations at various temperatures for n-GaN.

Fig. 4 shows the dependency of the low-field mobility of the n-GaN on the doping concentrations at room temperature using Equation (23). As seen from Fig. 4, at room temperature the electron mobility falls sharply as the doping in the n-GaN increases. This is again explained by the effect of increased ionized impurity scattering limiting electron drift mobility. Electron mobility saturates at $\mu_{max}(300) \cong 1572.3\ cm^2/V.s$ at very low impurity concentrations of $1x10^{15}$ cm^{-3}, while at very high doping concentrations of $1x10^{20}$ cm^{-3} it saturates at $\mu_{min}(300) \cong 125.275\ cm^2/V.s$. The obtained results are found to be in good agreement with those obtained by other researchers e.g. [22-23].

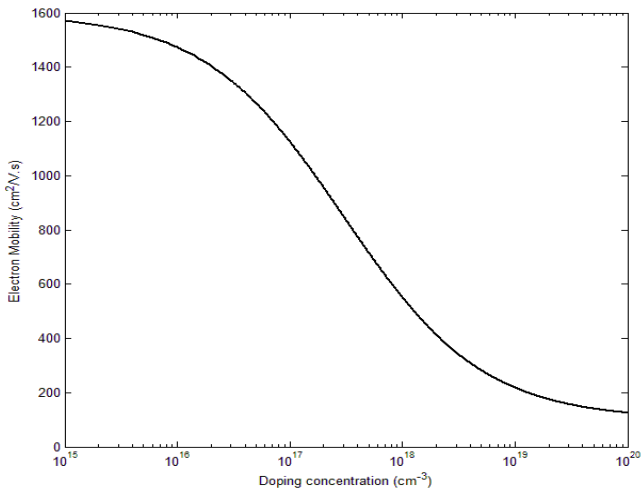

Fig. 4. Dependency of low-field electron mobility on doping concentrations at 300 K for n-GaN.

Fig. 5 represents the relation of the low-field mobility with changes of temperature at an impurity concentration of $1x10^{18}$ cm^{-3} for the n-GaN.

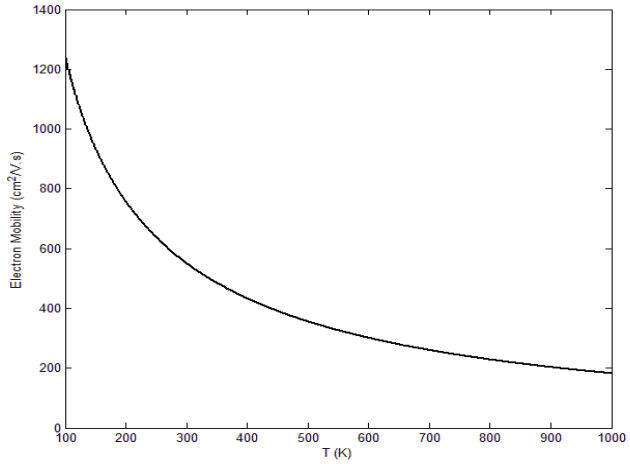

Fig. 5. Dependency of low-field mobility on temperature at a doping concentration of $1x10^{18}$ m^{-3} for n-GaN.

It is seen that the increase in temperature leads to a decrease in electron mobility. At room temperature, the mobility is 551.424 $cm^2/V.s$. The obtained results are found to be in good agreement with those found in the literatures; e.g. [23].

Such findings derived from these figures emphasize the necessity of accurately describing these effects for precise prediction of electron mobility in compound semiconductors.

IV CONCLUSION

Numerical results of electron mobility of the compound semiconductor n-GaN, due to some scattering mechanisms, show that the electron mobility of ionized impurities scattering of n-GaN increased as the temperature increased, whereas the mobility resulting from acoustic lattice vibrations decreased with increasing temperature. The mobility of piezoelectric potential scattering was found to be smaller than the mobility of deformation potential scattering. At higher temperatures, mobility value due to piezoelectric potential scattering was also near to that of deformation potential scattering suggesting that the effect of the piezoelectric acoustic phonons scattering may be ignored. The mobility attributable to the neutral impurity scattering was found to be constant since it is temperature independent. The drift mobility of n-GaN increased as temperature is raised until it reached a maximum value (at low temperatures), after which it decreases (at high temperatures). In addition, when the concentration of impurities increased, the peak value of mobility decreased. The drift mobility of the electrons of n-GaN decreased when the temperature was lowered, while the drift mobility increased when the impurity doping density decreased. Low-field electron mobility of n-GaN, at low temperatures, decreased sharply as the material doping increased explaining the effect of increased ionized impurity which limits the electron drift mobility at very low temperature. Low-field electron mobility decreased when the temperature increased.

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